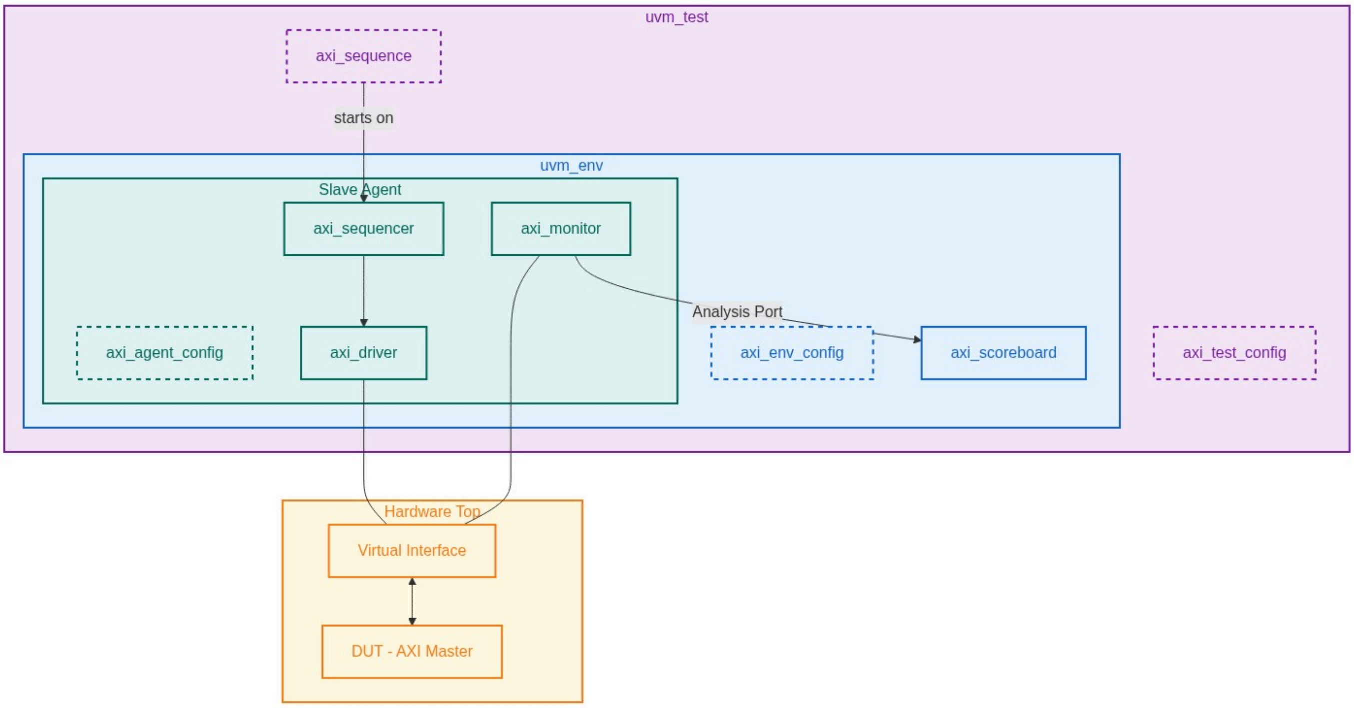
Verification Plan Report

Project: AXI-Stream Master VIP Verification Plan

Version: 4.0

Verification Role: Master

1. UVM Environment Architecture



2. Protocol Configurations

Feature	Configuration Name	Impact
Handshake Protocol	TVALID-Driven / TREADY Back-Pressure	The Slave Agent controls TREADY to inject back-pressure; the scoreboard checks that the Master (DUT) never drops TVALID or changes payload before handshake completes (Section 2.2, p.18).
Packet Framing	TLAST Boundary Generation	The Master must assert TLAST on the last beat of every packet. Test scope includes single-beat packets, multi-beat packets, and back-to-back packets (Section 2.6, p.26).
Byte Qualifiers	TKEEP / TSTRB Encoding	TKEEP marks null bytes (can be removed); TSTRB marks position bytes vs data bytes. The scoreboard validates the correct TKEEP/TSTRB combination per beat (Section 2.5, p.24).

Stream Identification	TID / TDEST Routing (Optional, Width-Parameterised)	If TID_WIDTH > 0, the Master may interleave streams. Coverage must bin each unique TID/TDEST pair and verify ordering is preserved (Section 2.7, p.27).
Sideband Signaling	TUSER Pass-through	TUSER is aligned per data byte. The monitor must preserve byte-association across width-conversion paths (Section 2.9, p.29).
Wakeup Signaling (AXI5)	TWAKEUP Pre-TVALID Assertion	TWAKEUP must be asserted before or with TVALID and must stay high until TREADY is seen. Impacts power-aware Slave agent sequencing (Section 2.3, p.20).
Parity Protection (AXI5)	Odd Parity - Check_Type: Odd_Parity_Byte_All	DUT (Master) must generate TDATACHK, TVALIDCHK, TLASTCHK, etc. on every beat. Scoreboard checkers validate parity using Table 5-1 rules (Chapter 5, p.44-48).
Reset Behavior	TVALID LOW during ARESETn	The Master must not assert TVALID until the rising ACLK edge after ARESETn goes HIGH. The driver monitors reset to flag violations (Section 2.8.2, p.28).
Continuous Packets Mode	Continuous_Packets = True	When enabled: no interleaving, no TSTRB, no mid-packet null bytes. TID/TDEST must remain stable while TLAST=0 (Section 3.3, p.38).

3. Test Requirements

Index	Name	Description
REQ_TR_01	TVALID Stability After Assertion	A Transmitter (Master DUT) is not permitted to wait until TREADY is asserted before asserting TVALID. Once TVALID is asserted, it must remain asserted and all payload signals (TDATA, TSTRB, TKEEP, TLAST, TID, TDEST, TUSER) must remain stable until the handshake occurs. Verification Intent: Inject random TREADY stalls via the Slave Agent driver. A protocol assertion in the monitor must fire if TVALID or any sideband changes before the handshake.

REQ_TR_02	Packet Boundary - TLAST Generation	TLAST marks the final beat of a packet. The number of TLAST assertions must be preserved end-to-end between Transmitter and Receiver. Merging of transfers with different TID/TDEST is not permitted once TLAST has been seen. Verification Intent: Verify the DUT correctly asserts TLAST on the last beat for single-beat packets, multi-beat packets, and back-to-back packets with no idle gaps.
REQ_TR_03	TKEEP / TSTRB Byte Qualifier Compliance	TKEEP=HIGH, TSTRB=HIGH = Data byte. TKEEP=HIGH, TSTRB=LOW = Position byte. TKEEP=LOW, TSTRB=LOW = Null byte. TKEEP=LOW, TSTRB=HIGH is RESERVED and must never be generated. Verification Intent: Randomize TKEEP/TSTRB patterns in sequences. An assertion must fire on the reserved combination. The scoreboard reconstructs the stream and checks data byte ordering.
REQ_TR_04	Reset Exit - TVALID After ARESETn	During ARESETn=LOW, the Master must drive TVALID=LOW. The first rising edge of TVALID is only valid at an ACLK edge following the rising edge of ARESETn. Verification Intent: Apply reset mid-transfer and verify the DUT deasserts TVALID immediately. Verify that transfers resume correctly only after ARESETn is deasserted and one clock passes.
REQ_TR_05	AXI5 Parity Generation on All Signals	When Check_Type=Odd_Parity_Byte_All (AXI5 only), the Master must generate odd parity on all signals each cycle that the Check Enable condition is met (e.g., TVALIDCHK every cycle ARESETn=1, TDATACHK every cycle TVALID=1). Parity must be driven for all bits, including inactive bytes (e.g., TKEEP=0 bytes still require correct TDATACHK). Verification Intent: Scoreboard checks all check signals on every valid cycle. Fault injection: force a bit-flip and check DUT does not assert a false parity.
REQ_TR_06	TWAKEUP Pre-Transfer Assertion (AXI5)	TWAKEUP must be asserted at least one cycle before TVALID. If TWAKEUP and TVALID are both HIGH in the same cycle, TWAKEUP must remain HIGH until TREADY is also HIGH. Verification Intent: Verify the DUT asserts TWAKEUP ahead of TVALID. Check that the Slave Agent (with power-aware mode) does not respond to TVALID until TWAKEUP is seen.

REQ_TR_07	TID/TDEST Transfer Ordering	The AXI-Stream protocol requires that all transfers remain ordered. Reordering is not permitted. Interleaving on a per-transfer basis is permitted between different TID/TDEST streams, but must not increase stream count beyond the Receiver's interleaving capability. Verification Intent: Monitor logs TID/TDEST per beat. The scoreboard checks in-order delivery for each unique stream identifier.
REQ_TR_08	Continuous Packets Constraint Mode	When Continuous_Packets=True: TSTRB must not be present; if TLAST=0, all TKEEP bits must be HIGH and TID/TDEST must not change. Null bytes are only permitted in the trailing bytes of the last beat (TLAST=1). Verification Intent: In continuous-packet test mode, a monitor assertion verifies TID/TDEST lock until TLAST=1 and that no mid-packet null bytes occur.

4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_001	Single-beat packet: Master asserts TVALID with TLAST=1 on the same cycle. Slave TREADY asserted after 0-4 random stall cycles. Verify data integrity and TVALID stability.	REQ_TR_01, REQ_TR_02
TC_002	Multi-beat packet (N=1 to 256 beats): TVALID held high across all beats. TREADY randomly stalled. TLAST asserted only on final beat. Scoreboard reconstructs and validates whole packet.	REQ_TR_01, REQ_TR_02
TC_003	Back-to-back packets: TVALID stays HIGH across TLAST boundary (packet N ends, packet N+1 starts next cycle). Verify no gap insertion and correct TLAST framing.	REQ_TR_02
TC_004	TKEEP/TSTRB sparse stream: Randomized TKEEP/TSTRB per beat, including null bytes (TKEEP=0). Assert that reserved combination TKEEP=0/TSTRB=1 never appears.	REQ_TR_03
TC_005	Null packet: TVALID=1, TLAST=1, all TKEEP=0 on first beat. Verify Master drives this correctly and Slave Agent acknowledges without data consumption.	REQ_TR_02, REQ_TR_03
TC_006	Mid-transfer reset: Assert ARESETn=LOW while TVALID=1, TREADY=1. Verify TVALID drops to 0 immediately. After deassertion, verify TVALID only rises on the next valid ACLK edge.	REQ_TR_04

TC_007	AXI5 Parity - normal traffic: All TVALIDCHK, TDATACHK, TLASTCHK signals verified on every qualifying cycle across 1000+ random beats.	REQ_TR_05
TC_008	AXI5 Parity - inactive bytes: TKEEP=0 on bytes [1] and [3] of 4-byte bus. TDATACHK must still be correct across all bytes including null byte positions.	REQ_TR_05
TC_009	TWAKEUP timing: TWAKEUP asserted exactly 1 cycle before TVALID. Verify Slave Agent waits for TWAKEUP before accepting the transfer.	REQ_TR_06
TC_010	Stream interleaving: Two TID streams (TID=0 and TID=1) interleaved transfer-by-transfer. Scoreboard reconstructs each stream independently and verifies ordering is preserved.	REQ_TR_07
TC_011	Continuous Packets mode: With Continuous_Packets=True, verify TID/TDEST do not change while TLAST=0, and TSTRB is absent from the interface.	REQ_TR_08

5. Functional Coverage Groups

Group Name	Description
cg_handshake_timing	Cross of TVALID raised cycles and TREADY latency (0 to 16 cycles). Ensures all handshake timing variants are exercised.
cg_packet_size	Bins: 1-beat, 2-4 beats, 5-16 beats, 17-64 beats, 65-256 beats. Includes back-to-back and isolated packet types.
cg_tkeep_tstrb_patterns	Walking 1s/0s and random combinations on TKEEP/TSTRB. Bins for: all-data, all-null, mixed, position-byte, reserved (must be 0-coverage).
cg_reset_scenarios	Bins: reset at idle, reset mid-packet (TVALID=1, TREADY=0), reset during handshake (both=1). Post-reset recovery verified.
cg_parity_coverage	TDATACHK validation across all byte lanes. Separate bins per byte offset, including zero-TKEEP bytes (inactive bytes that still need correct parity).
cg_twakeup_timing	Bins: TWAKEUP 0,1,2,4+ cycles before TVALID; simultaneous TWAKEUP+TVALID.
cg_stream_interleaving	Number of unique TID values: 1, 2, 4. Interleaving depth (consecutive transfers before switch) bins: 1, 2-8, 9+.

6. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input	1	Global clock. All signals sampled on rising edge (Section 2.8.1).

ARESETn	Input	1	Active-LOW reset. Asynchronous assert, synchronous deassert after rising ACLK (Section 2.8.2).
TVALID	Master Output	1	DUT asserts to indicate valid payload is being driven. Must remain high once asserted until TREADY is seen.
TREADY	Master Input	1	Slave Agent asserts to accept a transfer. Default HIGH if omitted. Used to inject back-pressure in test.
TDATA	Master Output	TDATA_WIDTH (must be integer x8)	Primary data payload. Must be stable while TVALID=1 and TREADY=0.
TSTRB	Master Output	TDATA_WIDTH /8	Per-byte: HIGH=data byte, LOW=position byte. Must be consistent with TKEEP per Table 2-3.
TKEEP	Master Output	TDATA_WIDTH /8	Per-byte: HIGH=must be delivered, LOW=null byte (can be removed). Default HIGH if signal absent.
TLAST	Master Output	1	Packet boundary marker. Asserted on last beat only. Number of assertions must be preserved end-to-end.
TID	Master Output	TID_WIDTH (rec. <= 8)	Stream identifier. Enables per-transfer interleaving when multiple streams share the interface.
TDEST	Master Output	TDEST_WIDTH (rec. <= 8)	Routing identifier. Coarse destination routing information for stream switching.
TUSER	Master Output	TUSER_WIDTH (rec. integer x TDATA_WIDTH /8)	User-defined sideband. Per-byte aligned. Not guaranteed when associated TKEEP=LOW.
TWAKEUP	Master Output	1	AXI5 only. Activity indicator. Must be asserted before TVALID. Requires Wakeup_Signal=True property.
TVALIDCHK	Master Output	1	AXI5 parity. Odd parity over TVALID. Driven every cycle where ARESETn=1.
TDATACHK	Master Output	TDATA_WIDTH /8	AXI5 parity. Odd parity per 8-bit lane of TDATA. Driven every cycle where TVALID=1, including null byte lanes.
TLASTCHK	Master Output	1	AXI5 parity. Odd parity over TLAST. Driven when TVALID=1.
TREADYCHK	Slave Output	1	AXI5 parity. Odd parity over TREADY. Driven by Slave Agent when ARESETn=1.

7. Verification Environment Structure

Root: axi_stream_master_vip_tb/

Verification Plan Report

|-- agents : Slave Agent (Driver as responder, Monitor as observer)

|-- env : UVM Environment (Scoreboard, Protocol Checker)

|-- sequences : Stimulus sequences: basic, sparse, back-to-back, reset, parity

|-- tests : UVM Test classes referencing each scenario

|-- top : TB top with virtual interface and DUT instance

|-- verif_plan : YAML plan, PDF report, and sub-prompts reference